



REVIEW ARTICLE

Clostridioides difficile infection in the allogeneic hematopoietic cell transplant recipient

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Clostridioides difficile (CD) is one of the most important causes of diarrhea in hospitalized patients, in particular those who undergo an allogeneic hematopoietic cell transplant (allo-HCT) and who are more at risk of developing a CD infection (CDI) due to frequent hospitalizations, iatrogenic immunosuppression, and prolonged antibiotic cycles. CDI may represent a severe condition in allo-HCT patients, increasing the length of hospitalization, influencing the intestinal microbiome with a bidirectional association with graft-versus-host disease, and leading to unfavorable outcomes, including death. The diagnosis of CDI requires the exclusion of other probable causes of diarrhea in HCT patients and is based on highly sensitive and highly specific tests to distinguish colonization from infection. In adult patients, fidaxomicin is recommended as first-line, with oral vancomycin as an alternative agent. Bezlotoxumab may be used to reduce the risk of recurrence. In pediatric patients, vancomycin and metronidazole are still suggested as first-line therapy, but fidaxomicin will probably become standard in pediatrics in the near future. Because of insufficient safety data, fecal microbiota transplantation is not routinely recommended in HCT in spite of promising results for the management of recurrences in other populations.

KEYWORDS

allogeneic hematopoietic cell transplantation, *Clostridioides difficile* infection, recurrent *Clostridioides difficile* infection, fecal microbiota transplantation

1 | INTRODUCTION

Clostridioides difficile infection (CDI) is the most common healthcare-associated infection in Europe and North America, posing a substantial

clinical burden in patients with hematologic diseases who are frequently hospitalized, immunosuppressed, and highly exposed to antibiotics.^{1,2} CDI is responsible for up to 10%–25% of antibiotic-associated diarrhea and 50%–75% of antibiotic-associated colitis.³ In patients < 18 years of age, *Clostridioides difficile* (CD) continues to be an underestimated cause of diarrhea because of the high rate of asymptomatic colonization (up to 70% in infants and 6% in children 1–2 years).⁴

Guidelines on the management of CDI in adults have recently been updated, introducing new approaches regarding therapy and diagnosis.^{5–7} In 2022, the American Society for Transplantation and Cellular Therapy (ASTCT) published practice guidelines for the diag-

List of Abbreviations: allo-HCT, allogeneic hematopoietic cell transplant; ASTCT, American Society for Transplantation and Cellular Therapy; auto-HCT, autologous hematopoietic cell transplant; CD, *Clostridioides difficile*; CDI, *Clostridioides difficile* infection; CI, confidence interval; EIA, enzyme immunoassays; ESCMID, European Society of Clinical Microbiology and Infectious Diseases; FMT, fecal microbiota transplantation; GDH, glutamate dehydrogenase; GvHD, graft-versus-host disease; HCT, hematopoietic cell transplant; ICU, intensive care unit; IDSA/SHEA, Infectious Diseases Society of America and Society for Healthcare Epidemiology of America; LOS, length of stay; MRT, microbiota restoration therapy; NAAT, nucleic acid amplification tests; PCR, protein chain reaction; rCDI, recurrent CDI; RR, risk ratio; SHR, sub-distribution hazard ratio.

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nosis and management of CDI in hematopoietic cell transplant (HCT) patients.⁸ Guidelines have also been published for pediatric cancer and HCT patients (now under revision by an international multidisciplinary group on infections in these patients).⁹

This review aims to explore the current understanding of CDI in pediatric and adult allogeneic HCT (allo-HCT), with a special focus on refractory and difficult-to-treat CDI. The roles of prevention and fecal microbiota transplant will also be discussed.

By summarizing the available evidence, we aim to provide a comprehensive overview of the challenges associated with CDI in HCT, highlighting recent advances in the field and identifying areas that require further investigation.

2 | EPIDEMIOLOGY, RISK FACTORS, AND OUTCOMES IN ALLO-HCT RECIPIENTS

2.1 | Adults

HCT recipients are especially at risk of developing CDI because of prolonged hospitalization, frequent broad-spectrum antibiotic therapy,¹⁰ damage to the intestinal mucosa caused by chemotherapy during conditioning regimens which causes impairment of the normal microbiome, and post-HCT immunosuppressive therapy that reduces the immune response against *C. difficile* and its toxins.^{8,11,12}

CDI usually occurs within the first month following allo-HCT; about 50% of all CDI cases occur after engraftment.^{13,14} Although many reports describe mild-to-moderate disease, CDI in HCT can be severe and associated with unfavorable outcomes. Many studies found no correlation between CDI and increased mortality in the HCT population.^{13,15,16} Nevertheless, other investigators have reported an association of CDI with mortality.^{17–21}

A bidirectional association between graft-versus-host disease (GvHD) and CDI is reported in the literature since both diseases are related to intestinal dysbiosis. GvHD has been found to be a risk factor for the development of CDI in HCT populations^{12,21–23} and a potential complication of CDI.^{18,19,22,23}

A review on CDI in HCT and hematologic malignancy populations conducted in 2019¹⁴ reported a 9-fold higher incidence among HCT patients than in the general population. In allo-HCT recipients, frequency is approximately 2-fold higher (14%–30.4%) compared with autologous transplants.²⁴ In a meta-analysis by Luo et al.,²⁵ the estimated prevalence of CDI was 15.3% in allo-HCT and 9.2% in autologous HCT (auto-HCT) recipients, and different incidence rates of CDI were found worldwide, with higher rates in North America (14.1%) and Asia (11.6%), and lower rates in Europe (10.7%).

To identify the incidence, risk factors, and outcomes of CDI between the start of conditioning and 100 days after HCT, Rosignoli et al.²⁶ conducted a retrospective study that included 261 allogeneic and 220 auto-HCT recipients. In the multivariate analysis, factors that were significantly associated with a higher risk of CDI development were previous abdominal surgery (sub-distribution hazard ratio [SHR]: 3.69; 95% confidence interval [CI]: 1.614–8.458), previous Hepatitis C Virus

infection (SHR: 6.53; 95% CI: 1.295–33.016), bacterial or fungal infections after HCT (SHR: 2.15; 95% CI: 0.044–0.529), and antibiotic therapy before HCT (SHR: 15.09; 95% CI: 6.491–35.078).

In a retrospective study on 466 allo-HCT and 1,454 solid organ transplant recipients,² patients with CDI had a longer length of stay (LOS) from transplant admission. For allo-HCT, the median LOS in days was 35 (range 19–141) for patients with CDI versus 29 (range 13–164) for patients without CDI.

The authors observed an association of CDI with severe (stages II–IV) acute GvHD within 30 days of HCT. Lower gastrointestinal GvHD within 100 days from transplant was more common among allo-HCT recipients with CDI (43.8% vs. 27.8%).² This study also showed that prior fluoroquinolone use was associated with an increased risk of CDI in multivariate analysis (HR 4.06; 95% CI: 1.31–12.63).

Weber et al.²⁷ performed a retrospective study evaluating the impact of CDI on HCT recipients. They included 276 patients receiving allo-HCT and 191 patients receiving auto-HCT. The peritransplant prevalence of CDI (from 30 days before to 100 days after HCT) was higher in the allo-HCT (17.6%) compared to the auto-HCT recipients (7.3%). In the allo-HCT cohort, overall survival and GvHD- and relapse-free survival between CDI and non-CDI patients were similar.

In the current study, GvHD was not associated with CDI in the multivariate analysis.²⁷

2.2 | Children

Rates of CDI are increasing over time in children,²⁸ particularly in the immunocompromised host most often exposed to risk factors, including recent antibiotic use, chemotherapy, and prolonged hospitalization.^{11,29–32} Children undergoing HCT are particularly vulnerable due to unique and non-modifiable risk factors,³³ but data on the incidence in this specific population are unfortunately quite scarce³⁴ and with some controversy.^{12,33,35–38} One study reported an 8% incidence, with higher rates in patients transplanted specifically for acute leukemia.³⁶ In another study, the incidence was 10% in the first 100 days post-HCT in pediatric patients receiving antibacterial prophylaxis, without significant correlation with primary disease, prior CDI or GvHD.³⁷

Recently, Ramanathan et al.³⁸ performed a study on 826 patients with CDI after allogeneic-HCT between the start of conditioning and day 100 and a control group of 6723 patients without CDI. In this study, the incidence of CDI was higher in pediatric patients (2–18 years) at 18.7% than in adult patients at 10.2%.

Risk factors for CDI specific to pediatric patients have been analyzed only recently. In children with cancer, admission for CDI was found to be associated with exposure to aminoglycosides, third- and fourth-generation cephalosporins, and proton pump inhibitors, as well as chemotherapy administration in the 8–14 days prior to admission.³⁰ Recently, Spruit et al. performed an in-depth HCT analysis on specific patterns of CDI; the overall rate of positive tests among HCT recipients during a transplant period was 14.7%, with a significant prevalence in those who received autologous compared to allo-HCT

(27.5% vs. 10.5%, $p < 0.001$). Interestingly, positive CDI tests were more related to a peripheral blood stem cell source than either bone marrow or umbilical blood sources (28.4% vs. 10.1% and 9.4% respectively, $p < 0.001$). Moreover, risk factors such as the use of corticosteroids, neutropenia, proton pump inhibitors or H_2 blockers, and cefepime or vancomycin exposure were found to be insignificant.³³ In another retrospective study on 109 cases of CDI in oncological pediatric patients, 29% of CDI cases were observed in HCT recipients. Quite surprisingly, CDI was associated with auto-HCT but not with allo-HCT or standard chemotherapy. Notably, allo-HCT recipients received ciprofloxacin prophylaxis after discharge from the transplant procedures, probably influencing the prevalence of CDI in this population.³⁹ This is suggested by an observation that reported lower CDI incidence in the case of fluoroquinolone prophylaxis, maybe due to lower therapeutic antimicrobial exposure in the prophylactic group.⁴⁰ On the other hand, during auto-HCT, an intensive preparatory regimen was administered with consequent more severe mucositis. This finding on allogeneic versus autologous transplant is in contrast with the data on CDI in adults, which reported a higher overall incidence in allogeneic (12%–27%) compared to auto-HCT (6%–8%),⁴¹ and it is also in contrast with recent pediatric data in HCT (8.9% allo-HCT vs. 6.7% auto-HCT).⁴²

In general, although CDI in children is frequently mild, in those receiving antineoplastic chemotherapy or HCT, severe or complicated CDI is not uncommon and is associated with prolonged LOS and an increased risk of death.³⁰ Age < 5 years and an underlying disease other than acute leukemia seem to be positive prognostic factors influencing the clinical course of CDI.⁴² A recent multicenter, multinational, retrospective study in patients with cancer and HCT recipients reported a 9.7% incidence of severe CDI episodes even though HCT was not found to be one of the factors significantly associated with this clinical picture.⁴³ Another single-center study did not report significantly higher rates of CDI-related complications in children with HCT, likely related to the small number of complications in the study, though these patients were more likely to require an intensive care unit (ICU) admission related to CDI (2.4% vs. 2%, $p = 0.84$).³⁴ Finally, other authors observed that CDI after auto-HCT was associated with a delay in further chemotherapy and prolonged hospitalization, in the absence of increased morbidity, ICU admissions and mortality.^{39,44}

3 | DIAGNOSIS AND SEVERITY ASSESSMENT

3.1 | Adults

The latest guidelines on diagnosis^{7,46} recommend testing for *C. difficile* only in symptomatic patients with three or more unformed stools in 24 h. CDI can also be suspected with an acute worsening of chronic diarrhea and may be accompanied by abdominal cramping/pain, nausea, vomiting, and fever. Severe CDI is defined by the presence of leukocytosis, fever (38.5°C or higher), or rise in serum creatinine (>1.5 mg/dl or $>50\%$ above baseline).^{5,6} The severity of CDI in HCT patients cannot always be evaluated using the same scoring systems as those used in

the general population^{47–50} because some predictors of severity (such as increased white blood cell count and fever) may be absent. ASTCT suggest testing a patient after evaluating diarrhea with a comprehensive review of medication and removing any potentially offending agents.⁸

When CDI is strongly suspected, it is important to use the proper diagnostic method. It is worth noticing that 11%–39% of allo-HCT patients are expected to be colonized with toxigenic *C. difficile* before transplant,⁵¹ and this can increase the risk of early CDI but can also lead to overdiagnosis in patients presenting symptoms of diarrhea but only being colonized by and not infected with a toxigenic *C. difficile* strain.^{8,52}

The most frequently used laboratory tests for *C. difficile* diagnosis are glutamate dehydrogenase (GDH) assay, enzyme immunoassays (EIAs) for toxins A and B, and nucleic acid amplification-protein chain reaction tests (NAAT-PCR).

The GDH assay that detects an enzyme present in toxigenic and nontoxigenic isolates is highly sensitive but not specific and is usually used in combination with a NAAT test or toxin EIA. The EIA detects toxins A and B. Toxin EIAs are also not optimal because of their sensitivity range (67%–92%).¹⁴ The NAAT-PCR test detects the gene for toxin B rather than its product. In contrast with the toxin EIAs, NAAT-PCR is highly sensitive and has a high negative predictive value for CDI. However, because of its high sensitivity, it has a poor positive predictive value for CDI.

A multistep algorithm for diagnosis might include, for example, NAAT plus EIA, or GDH and EIA. In the case of inconsistent results between GDH and toxin EIA tests, NAAT has to be performed to settle discrepancies.^{6–8}

3.2 | Children

The diagnosis of CDI in children does not differ from that in adults. Severity assessment in children with CDI is a challenging problem, and, especially in cancer and HCT populations, it is an important research gap because clear definitions for non-severe and severe CDI have not been established. On the basis of clinical experience, Diorio et al., in a document with clinical practice guidelines for the management of CDI in pediatric HCT recipients, proposed as a provisory definition of severe CDI the presence of toxic megacolon, pseudomembranous colitis or hemodynamic instability.⁹ However, it is possible that pseudomembranes may not be present in the neutropenic patient because the pathogenesis of this complication relies on neutrophil infiltrates, just as neutrophil leukocytosis is absent in chemotherapy-induced neutropenia.^{34,53} The absence of a definition of severe disease means that recommendations for therapy management often specifically refer to non-severe disease.⁹

More recently, Haeusler et al., in a multicenter retrospective study including pediatric patients with cancer and HCT recipients, proposed a definition of severe CDI as the presence of colitis, pneumatosis intestinalis, pseudomembranous colitis, ileus, or surgery for CDI, which occurred in 9.7% of patients; this definition is composed of



microbiological, clinical and radiographic criteria, it is pediatric-specific, and it takes into account the peculiar characteristics of patients with cancer or HCT.⁴³

These proposed definitions can be the starting point for further studies that could fill the gap to provide a definition of severe disease in children in general, particularly in the pediatric population undergoing bone marrow transplantation.

4 | TREATMENT

4.1 | Adults

4.1.1 | Initial episode of CDI

In the last few years, multiple studies have shown the non-inferiority of fidaxomicin over oral vancomycin for the treatment of an initial episode of CDI.^{5,54,55} Furthermore, fidaxomicin causes less disruption to the gut microbiome thanks to its restricted spectrum of activity^{56–58} and decreases recurrences compared with vancomycin.^{55,59} Updated guidelines for the treatment of a first episode of CDI by the Infectious Diseases Society of America and Society for Healthcare Epidemiology of America (IDSA/SHEA), the European Society of Clinical Microbiology and Infectious Diseases (ESCMID),^{5,6} and ASTCT⁸ suggest using fidaxomicin for 10 days in adults with an initial episode of CDI, with vancomycin becoming an alternative regimen. These guidelines agree on the benefit of interruption (or de-escalation) of antibiotic therapy and proton pump inhibitors. Metronidazole is no longer recommended as a first-line agent (See Table 1).^{48,60}

For patients with fulminant or severe-complicated CDI, defined as CDI in the presence of hypotension, septic shock, elevated serum lactate, ileus, toxic megacolon, bowel perforation, or any fulminant course of disease (i.e., rapid deterioration of the patient), a regimen of vancomycin 500 mg four times daily by mouth or nasogastric tube in addition to intravenous metronidazole (500 mg every 8 h) should be administered. For patients with an ileus, consider adding a rectal instillation of vancomycin if not contraindicated.

There is still no data to support the use of fidaxomicin over vancomycin in severe and severe-complicated CDI.^{5–8} Lastly, in fulminant CDI, tigecycline can be used as an intravenous agent,⁵ although there is only one case of its use described for an HCT recipient in 2014 for refractory CDI.⁶¹

Refractory CDI is defined as CDI not responding to recommended CDI antibiotic treatment after 3–5 days of therapy.⁵ For patients with non-complicated CDI, who are not responding to standard-of-care treatment, are in a non-deteriorating condition, and are not progressing to complicated CDI, the diagnosis should be reconsidered, as vancomycin and fidaxomicin resistance is very rare in Europe.⁵ In a pan-European study that included 918 *C. difficile* isolates, the susceptibility percentages for vancomycin and fidaxomicin were 96.8% and 100%, respectively.⁶² ASTCT suggests considering other diagnoses, such as gastrointestinal-GvHD or infectious colitis, in patients not responding as expected to CDI therapy.⁸

The use of bezlotoxumab is suggested by updated guidelines^{5–8} for patients at risk of recurrence. Risk factors for recurrence are defined as older age (>65 years), patients with prior CDI episode(s), patients with healthcare-associated CDI and hospitalization in the last three months, patients with concomitant non-CDI antibiotic use after the diagnosis of CDI, patients using proton pump inhibitors started during/after CDI diagnosis, patients with compromised immunity, and patients with a CDI due to virulent ribotypes.^{5,6}

Bezlotoxumab is a human-derived monoclonal antibody that binds to toxin B of *C. difficile*, neutralizing its activity. The efficacy of this molecule in preventing recurrent CDI has been studied with two phase III trials^{63,64} that enrolled a total of 1,554 patients (bezlotoxumab: 781; placebo: 773) between November 2011 and May 2015 (MODIFY I and II). The pooled analysis demonstrates that the addition of bezlotoxumab reduced CDI recurrence after initial clinical treatment at 12 weeks (risk ratio [RR]: 0.62; 95% CI: 0.51–0.75) and reduced CDI-associated hospital readmission at 30 days (RR: 0.46; 95% CI: 0.29–0.71). In patients with a history of congestive heart failure, bezlotoxumab should be reserved for use when the benefit outweighs the risk.^{5,6}

4.2 | Treatment of a first recurrence

Recurrent CDI (rCDI) is defined by ASTCT guidelines as a new onset of symptoms suggestive of CDI, with a positive assay after a successfully treated first episode, with a time of onset of up to 4–12 weeks after the end of the prior treatment,^{8,47} while ESCMID guidelines consider an episode of CDI to be recurrent if it occurs within 8 weeks from a previous episode.⁵

In HCT recipients, fidaxomicin is the recommended first-line therapy, regardless of which agent was used for the first episode, with a 10-day treatment course of oral fidaxomicin 200 mg twice daily, or an extended course of fidaxomicin with 200 mg twice daily for the first 5 days and 200 mg once daily for another 20 days.⁸ If fidaxomicin is not available, an acceptable alternative is oral vancomycin with a taper-pulse regimen, although tapers have not been standardized in duration.⁸

Fewer recurrences were found with vancomycin taper ($n = 29$, recurrence rate 31%) or pulse ($n = 7$, 14%) than with standard vancomycin treatment for 1–2 weeks ($n = 83$, 43–71%) in a secondary analysis of the placebo arm of two large RCTs.⁶⁵

The tapering scheme proposed by ESCMID and ASTCT is 2 weeks of oral vancomycin 125 mg four times daily, followed by 1 week of 125 mg twice daily, then 1 week of 125 mg daily, then 1 week of 125 mg every second day, and finally 125 mg every third day for 1 week.^{5,8,66}

ASTCT guidelines suggest using bezlotoxumab to prevent a second rCDI in patients who were not treated with bezlotoxumab during the first episode.^{8,63} Bezlotoxumab has been seen to reduce recurrence in patients with one prior episode by 13.9% and in patients with two or more prior episodes by 15.7% (See Table 1).^{63,64}

**TABLE 1** Summary of updated practical guidelines for the management of *Clostridium difficile* infection (CDI).

	Initial CDI episode	First CDI recurrence	Second CDI recurrence	Fulminant CDI
ASTCT (8) (2022)	Definition: Unexplained, new-onset diarrhea (3 or > loose bowel movements in a 24-h period) or acute worsening of chronic diarrhea, may be accompanied by abdominal cramping/pain, nausea, vomiting, and fever Preferred: Fidaxomicin 200 mg bid x 10 days OR vancomycin 125 mg qid x 10 days Alternative: Metronidazole PO (adults): 500 mg tid (if VAN or FDX are not available) x 10 days IV Adjunctive: Bezlotoxumab as a single intravenous infusion (10 mg/kg)	Definition: A new episode of symptom onset consistent with CDI and a positive assay after a successfully treated episode of CDI with an onset ranging from within days to up to 4–12 weeks after cessation of treatment for the prior episode Preferred: Fidaxomicin 200 mg bid x 10 days or an extended course of fidaxomicin (200 mg bid x 1–5 days, then every other day x days 7–25) Alternative: Oral vancomycin 125 mg qid for the first week + bid for a week + qd for a week + every 48 h for a week Adjunctive: If not previously administered, bezlotoxumab, given as a single intravenous infusion (10 mg/kg)	Definition: N/A Preferred: Fidaxomicin 200 mg bid x 10 days OR extended dosing of fidaxomicin 200 mg bid x 1–5 days, then every other day for days 7–25) OR taper-pulse of fidaxomicin (for example: 200 mg bid x 10 days, qd for 7 days, then once every other day for 26 days [total 40 capsules]) Alternative: Oral vancomycin 125 mg qid for the first week + bid for a week + qd for a week + every 48 h for a week Adjunctive: If not previously administered, bezlotoxumab, given as a single intravenous infusion (10 mg/kg)	Definition: CDI associated with hypotension or shock, ileus, or megacolon Preferred: Oral vancomycin 500 mg qid by mouth or by nasogastric tube Alternative: N/A Adjunctive: For whom there is no contraindication to rectal installation, consider adding a rectal installation of vancomycin. Metronidazole 500 mg tid IV
ESCMD (5) (2021)	Definition: Clinical findings compatible with CDI and microbiological evidence of <i>C. difficile</i>-free toxins by enzyme immunoassay without reasonable evidence of another cause of diarrhea OR a clinical picture compatible with CDI and a positive nucleic acid amplification test preferably with a low cycle threshold value, or positive toxigenic <i>C. difficile</i> culture OR pseudomembranous colitis as diagnosed during endoscopy, after colectomy or on autopsy, in combination with a positive test for the presence of toxigenic <i>C. difficile</i> Preferred: Fidaxomicin 200 mg bid x 10 days OR Vancomycin 125 mg qid x 10 days Alternative: Metronidazole PO (adults): 500 mg tid (if VAN or FDX are not available) x 10 days Adjunctive: Bezlotoxumab as a single intravenous infusion (10 mg/kg) in high risk of recurrence	Definition: When CDI recurs within 8 weeks after a previous episode, provided the symptoms from the previous episode resolved after completion of initial treatment Preferred: SoC + Bezlotoxumab OR Fidaxomicin 200 mg bid x 10 days OR consider extended course of fidaxomicin (200 mg bid x 1–5 days, then every other day x days 7–25) Alternative: Oral vancomycin 125 mg qid for the first week + bid for a week + qd for a week + every 48 h for a week	Definition: N/A Preferred: Fecal microbiota transplantation OR SoC + Bezlotoxumab Alternative: Oral vancomycin 125 mg qid for the first week + bid for a week + qd for a week + every 48 h for a week	Definition: CDI presenting one of the following: hypotension, septic shock, elevated serum lactate, ileus, toxic megacolon, bowel perforation or any fulminant course of disease (i.e., rapid deterioration of the patient) Preferred: Fidaxomicin 200 mg bid x 10 days OR vancomycin 125 mg qid x 10 days + multidisciplinary approach with surgical consultation Alternative: N/A Adjunctive: Rectal or nasoduodenal vancomycin. Consider IV metronidazole or tygecilin

(Continues)



TABLE 1 (Continued)

	Initial CDI episode	First CDI recurrence	Second CDI recurrence	Fulminant CDI
IDSA/SHEA (6) (2021)	Definition: N/A Preferred: Fidaxomicin 200 mg bid x 10 days Alternative: Oral vancomycin 125 mg qid x 10 days If above not available oral metronidazole 500 mg tid x 10–14 days	Definition: N/A Preferred: Fidaxomicin 200 mg bid x 10 days, OR bid x 5 days followed by once every other day for 20 days Alternative: Oral vancomycin 125 mg qid x 10 days. Consider a standard course of vancomycin if metronidazole was used for treatment of the first episode. Adjunctive: Bezlotoxumab 10 mg/kg given intravenously once during administration of SoC antibiotics	Definition: N/A Preferred: Fidaxomicin 200 mg bid x 10 days, OR bid x 5 days followed by once every other day for 20 days OR oral vancomycin in a tapered and pulsed regimen OR oral vancomycin 125 mg qid x 10 days followed by rifaximin 400 mg tid x 20 days Fecal microbiota transplantation Alternative: N/A Adjunctive: Bezlotoxumab 10 mg/kg given intravenously once during administration of SOC antibiotics	Definition: CDI presenting hypotension or shock, ileus, megacolon Preferred: Oral vancomycin 500 mg qid by mouth or by nasogastric tube. If ileus, consider adding rectal instillation of vancomycin. Alternative: N/A Adjunctive: IV metronidazole 500 mg tid should be administered together with oral or rectal vancomycin, particularly if ileus is present.
ACG (7) (2021)	Definition: N/A Preferred: Vancomycin qid x 10 days OR fidaxomicin 200 mg bid x 10 days Alternative: Oral metronidazole tid x 10 days Adjunctive: Bezlotoxumab as a single intravenous infusion (10 mg/kg) in high risk of recurrence	Definition: The recurrence of diarrhea and a confirmatory positive test (NAAT or EIA) within 8 weeks after treatment of an initial episode of CDI Preferred: Tapering/pulsed-dose vancomycin for patients experiencing a first recurrence after an initial course of fidaxomicin, vancomycin, or metronidazole OR fidaxomicin for patients experiencing a first recurrence after an initial course of vancomycin or metronidazole Adjunctive: Bezlotoxumab as a single intravenous infusion (10 mg/kg) in high risk of recurrence	Definition: N/A Preferred: Fecal microbiota transplantation Alternative: For patients with rCDI who are not candidates for FMT, who relapsed after FMT, or who require ongoing or frequent courses of antibiotics, long-term suppressive oral vancomycin at a dose of 125 mg qd for a minimum of 8 weeks Adjunctive: Bezlotoxumab as a single intravenous infusion (10 mg/kg) Oral vancomycin prophylaxis may be considered during subsequent systemic antibiotic use in patients with a history of CDI who are at high risk of recurrence to prevent further recurrence.	Definition: CDI presenting hypotension or shock, ileus, megacolon Preferred: Vancomycin 500 mg qid for the first 48–72 h, for patients with ileus, the addition of vancomycin enemas (500 mg qid) + multidisciplinary approach with surgical consultation Alternative: Fecal microbiota transplantation for patients with fulminant CDI refractory to antibiotic therapy, in particular when patients are deemed poor surgical candidates Adjunctive: IV metronidazole 500 mg tid
Pediatric International Guidelines (9) (2018)	Preferred: Use either oral metronidazole or oral vancomycin for the treatment of nonsevere CDI; Use oral vancomycin for the treatment of severe CDI	Preferred: Consider fidaxomicin for the treatment of recurrent CDI	Adjunctive suggests that monoclonal antibodies not be used routinely for the treatment of CDI Do not use fecal microbiota transplantation routinely for the treatment of CDI	

Abbreviations: ACG, American College of Gastroenterology; ASTCT, American Society for Transplantation and Cellular Therapy; bid, bis in die; CDI, *Clostridioides difficile* infection; EIA, enzyme immunoassay; ESCMID, European Society of Clinical Microbiology and Infectious Diseases; FDX, fidaxomicin; IDSA/SHEA, Infectious Diseases Society of America/Society for Healthcare Epidemiology of America; IV, intravenous; NAAT, nucleic acid amplification test; qid, quaque diem; SoC, standard of care; tid, ter in die; VAN, vancomycin.

4.3 | Treatment of multiple recurrences

Management of a second or greater rCDI can be challenging, and the optimal therapy has not been defined. The therapeutic regimen proposed by ASTCT guidelines are standard dosing of fidaxomicin for 10 days or extended dosing of fidaxomicin (200 mg twice daily for 1–5 days, then every other day until day 25) or taper-pulse of fidaxomicin (for example, 200 mg twice daily for 10 days, once per day for 7 days, then 200 mg every 48 h for 26 days for a total of 40 capsules). This scheme is based on a single-center retrospective case series with a total of 46 patients (seven of them immunosuppressed) with multiple rCDI, treated using tapered-pulse regimen fidaxomicin. All patients were initially given a treatment course of either oral vancomycin four times daily or fidaxomicin twice daily. Sustained clinical response rates at 30 and 90 days were 74% (34/46) and 61% (28/46), with 2 patients out of 46 developing new CDI symptoms after starting the tapering-pulse fidaxomicin course. Other approaches proposed are taper-pulse of oral vancomycin and fecal microbiota transplantation (FMT).

In three studies published between 2019 and 2021, bezlotoxumab resolved rCDI in over 71% of patients,^{67–69} and it can be given for multiple recurrences if not previously administered.

4.4 | Fecal microbiota transplantation

In patients who have experienced multiple recurrences of CDI, treatment can be challenging due to a recurrence risk higher than 60%.⁷⁰ The higher risk of recurrence is related to chronic microbiota dysbiosis, which has not been observed in individuals who recovered from CDI.^{8,71} For this reason, microbiota restoration therapy (MRT) or FMT are potentially valuable options for the treatment of rCDI.^{72,73}

Multiple standardized microbiome-based therapies have been studied for the prevention of rCDI in populations that did not include immunosuppressed patients (reviewed⁷⁴), and various case reports and retrospective studies have been published reporting good outcomes using FMT in HCT and immunosuppressed patients.^{75–78} The first case report of FMT performed on an allo-HCT recipient was published by Neemann et al. in 2012⁷⁵ for the treatment of fulminant CDI with a successful outcome. A retrospective study was published in 2017⁷⁶ to evaluate the safety and efficacy of FMT for the treatment of rCDI in patients with cancer treated with cytotoxic chemotherapy. It included 23 patients with hematologic¹³ and solid organ¹⁰ malignancies who underwent a single infusion of donor stool via colonoscopy. Of the 12 evaluable patients at 60 days follow-up, 11 (84.6%) achieved resolution of the diarrhea. There were no other severe adverse events and no infectious complications directly attributable to FMT.

Webb et al. described seven allo-HCT recipients who underwent FMT via a nasojejunal tube or colonoscopy from an unrelated donor. All of them obtained remission, with six patients having no relapse and one who needed another FMT; no donor-derived infections were reported.⁷⁷

In a study by Moss et al., eight patients received FMT as oral encapsulated therapy from unrelated donors, and all of them achieved resolution of the rCDI at 8 weeks. One of them experienced recurrence 179 days post-FMT. No serious adverse events attributable to FMT were reported.⁷⁸

In 2019, Defilipp et al. described two donor-derived *Escherichia coli* extended-spectrum beta-lactamase infections resulting in death in two FMT recipients.⁷⁹ The two patients received FMT from the same unrelated donor; one of them was an allo-HCT patient who was enrolled in a phase 2 trial to administer preemptively FMT oral capsules before and after allogeneic hematopoietic-cell transplantation (NCT03720392), and who died 5 days after allo-HCT and 8 days after the last FMT dose.

Currently, a randomized, open-label, controlled trial to assess the efficacy of autologous FMT in CDI prevention in allo-HCT recipients is underway (NCT02269150). The study will compare an arm of patients who will be observed and treated as per the standard of care and an arm of patients treated with FMT using feces stored prior to the initiation of the conditioning regimen. Fecal stools are analyzed by 16S rRNA gene sequencing and tested by assay for intestinal pathogens, including *C. difficile*, to study the microbiome of patients prior to and after the conditioning regimen and allo-HCT in both arms.

Although there are still no sufficient safety and tolerability data to recommend MRT/FMT for the treatment of rCDI in HCT patients, what has been published so far is promising. However, at the present time, the choice of this procedure should be tailored to the single patient, being careful to assess the risks and benefits.

4.5 | New approaches

New strategies for the prevention and treatment of CDI are under investigation; one of them is the preservation of the gut microbiome reducing the impact of antibiotics on the lower gastrointestinal tract. Ribaxamase is an oral enteric-coated beta-lactamase pellet that is intended to be used with intravenous penicillins and cephalosporins to prevent the disruption of the microbiome. It acts by degrading beta-lactam antibiotics in the lower gastrointestinal tract.⁸⁰ Ribaxamase has been tested in a double-blind, phase 2b, randomized placebo-controlled trial on 433 patients receiving intravenous cephalosporins. It has shown an incidence of primary CDI of 1% in the ribaxamase group and 3.4% in the placebo group.⁸¹

DAV132 is an oral-activated charcoal that exhibits release in the distal ileum. It has been tested on 44 volunteers and is shown to decrease free moxifloxacin fecal concentrations by 99%, preserving the intestinal microbiome.⁸²

Another strategy in the prevention of primary CDI that is being explored is vaccination.^{83,84} CDIFFENCE is a toxin-based vaccine that has been tested in a randomized, placebo-controlled, two-stage, phase 2 study and has shown a rate of seroconversion for toxin A of 92.3% and for toxin B of 53% at day 180.⁸⁵

VLA84 is a recombinant fusion protein comprising epitopes of both toxin A and toxin B that has completed its phase II trial (NCT02316470), showing a long-term immune response in adults.⁸⁶

Toxin-based vaccines do not prevent CD colonization and cannot restrain the spread of spores from the host to the environment, resulting in a possible increase of asymptomatic carriers.^{87,88} For this reason, new passive immunotherapies that target different pathogenic factors, such as pilin, spore proteins, flagellum, and S-layer proteins, are being explored.^{89–91}

4.6 | Probiotics

The use of probiotics is not currently recommended for the prevention of CDI or rCDI in HCT patients.⁸ Recently, Goldenberg et al. performed a meta-analysis to assess the efficacy and safety of probiotics for preventing CDI. They found that probiotics have shown to be effective in trials with an estimated risk of CDI >5%, but the interpretation of the results is complicated because the trials that were included in the meta-analysis used different probiotic formulations.^{5,92} Furthermore, cases of invasive infections after oral administration of probiotics in immunocompromised patients have been described,^{93,94} suggesting that bacteria and fungi contained in these formulations can lead to disseminated infections.⁹⁵

4.7 | Children

The 2017 IDSA/SHEA *C. difficile* guidelines recommended the same treatment approach for adults and pediatrics.⁴⁷ However, in 2021 they published a focused update, restricted to adults, which included new data and recommendations for fidaxomicin and bezlotoxumab.⁶

Clinical practice guidelines for the management of CDI in children and adolescents with cancer and undergoing HCT recommend using either oral metronidazole or oral vancomycin for the initial treatment of non-severe CDI because of similar cure and recurrence rates for both antibiotics.⁹ Vancomycin is recommended as first-line because of better palatability and fewer drug interactions since it is not absorbed systemically. In 2020, fidaxomicin was approved by the Food and Drug Administration and European Medicines Agency for use in CDI infections in children after the publication of two studies. The first was a phase 2 study on 38 patients <18 years old that reported good results in terms of efficacy (response rate 92%) and safety. The second study reports the results of a multicenter, randomized, single-blind phase 3 trial (SUNSHINE): compared with vancomycin, fidaxomicin was found to be better tolerated and with a higher global cure rate in children and adolescents with CDI.^{96,97} Of 142 patients treated, the emergent adverse events were similar for vancomycin and fidaxomicin, whereas the global cure at the end of treatment was significantly higher in participants receiving fidaxomicin (68% vs. 50%; adjusted treatment difference 18.8%, 95% CI: 1.5%–35.3%).⁹⁶ The main limitations to the use of fidaxomicin remain the cost and unavailability of the oral suspension. Even though further studies are needed, especially in allo-HCT patients, it can be reasonably assumed that recommendations will be changed in the next revision.

Only a few pediatric studies report the incidence and risk factors of rCDI in HCT. In a retrospective analysis on allo-HCT in patients aged 2–

73, unfortunately not stratified by age, the only significant risk factor for recurrence at multivariate analysis was the number of antecedent antibiotics (HR 1.96%; 95% CI: 1.09–3.52, $p = 0.025$). Recurrence was also associated with a trend of increased risk of mortality.²⁰ In one specific pediatric study, the rate of recurrence was not significantly higher in HCT compared with no transplant (29% vs. 21%, $p = 0.366$), even when the most important risk factor for rCDI was a diagnosis of malignancy requiring HCT (adjusted odds ratio 3.39; 95% CI).^{34,45} In another series, rCDI was observed in 20.7% of HCT pediatric patients (HR = 0.6; 95% CI: 0.3–1.7; $p = 0.8$)⁴² and in a study comparing HCT/cancer patients with subjects with other hematologic diseases or other pediatric populations, rCDI was significantly more frequent in the former group (14.4% vs. 5.3% and 2.8%, $p < 0.001$).³³ Finally, a pediatric study in hospitalized and outpatient pediatric HCT reported 22% rCDI, and the use of H₂ antagonists was found to be an independent risk factor for rCDI.⁴⁴

As for the use of bezlotoxumab to prevent rCDI, very recently the results of a multicenter, double-blind, placebo-controlled trial of bezlotoxumab administration in children receiving antibacterial therapy for CDI (MODIFY III trial) have been published. Overall, 143 patients were treated (107 with bezlotoxumab and 36 with placebo); bezlotoxumab at a dosage of 10 mg/kg was generally well-tolerated with placebo-like side effects and no drug discontinuation for these effects. Recurrences of CDI were low and comparable for bezlotoxumab and placebo (11.2% vs. 14.7%). However, this trial was designed to study the pharmacokinetics and safety of the monoclonal antibody in children, not to demonstrate a difference in CDI recurrence (because of the sample size limitation due to low incidence of CDI in pediatrics).⁹⁸ Another consideration is that the high rate of initial clinical response and low rate of recurrence may reflect the changing epidemiology of pediatric CDI and increased use of vancomycin and fidaxomicin.^{43,99} Notably, about two-thirds of enrolled patients had immunosuppression as a risk factor, including pediatric HCT patients. Unfortunately, there is no stratification of immunosuppressed patients to evaluate the sample number of this specific population.

The administration of a probiotic in addition to standard treatment of CDI in children is not currently recommended.⁹ Even though available trials show a significant reduction in recurrences,^{100–102} these trials do not include immunocompromised patients. The use of probiotics has been associated with cases of *Saccharomyces* fungemia, *Lactobacillus* bacteremia, and absidiomycosis.^{93,94}

As for FMT, there is a strong recommendation against its routine use in children in general and particularly in those with cancer or HCT. At present there is only one mixed-age trial available, with 3 children included, but chemotherapy was an exclusion criterion. Pediatric patients with cancer or HCT should be included in future research that evaluates FMT for the treatment of CDI.⁹

4.8 | Primary and secondary prophylaxis

The role of primary and secondary prophylaxis of CDI is still debated, and the data are not sufficient to recommend the use

of any agent for prophylaxis in HCT patients according to ASTCT guidelines.⁸

The use of vancomycin as a primary prophylaxis has been seen to reduce the incidence of CDI in one retrospective study on adult HCT recipients conducted by Ganetsky et al. in 2019;¹⁰³ nonetheless, there are no randomized control trials on HCT that validate this result. Moreover, vancomycin leads to disruption of the microbiome, which has been associated with worse outcomes in allo-HCT.^{104,105}

Prophylactic fidaxomicin may be a more acceptable option for CDI prevention. It is less disruptive to the normal gut microbiota compared to vancomycin⁹⁷ and spares commensal enteric bacteria such as *Bacteroides* spp.¹⁰⁶

In a double-blind randomized controlled study, Mullane et al. in 2019 evaluated the use of fidaxomicin versus placebo for the primary prevention of CDI in adults undergoing HCT.¹⁰⁷ Prophylaxis failure was similar in fidaxomicin and placebo recipients (28.6% vs. 30.8%), so the study did not meet its primary composite endpoint, which was prophylaxis failure through 30 days after discontinuation of the drug. Furthermore, prophylaxis with oral fidaxomicin can promote resistance to this agent;¹⁰⁸ therefore, data are insufficient to recommend it.

Lastly, the use of metronidazole is not recommended because of the risk of toxicity and lack of efficacy.^{109,110}

A study of patients who underwent allo-HCT found that the probability for developing post-HCT CDI during the peri-transplant period (28 days after transplant) was 39% in those who developed CDI within nine months before transplant, providing a rationale for secondary prevention to be investigated in this group of patients.¹¹¹

One last note should be added regarding prevention. Since the hands of the health care providers and the furnishings and materials in the room can harbor and transmit *C. difficile* spores, in order to prevent the spread of the infection, it is absolutely necessary to scrupulously comply with the correct, specific rules of hand hygiene and environmental cleaning, and the isolation standards required to control the spread of this infection.¹¹²

5 | SUMMARY AND CONCLUSION

Patients undergoing allo-HCT are especially at risk of developing CDI and new CDI recurrences because of unique risk factors, such as chemotherapy, repeated and prolonged hospitalization, and antibiotic treatments, that cause impairment of the normal microbiome.

The diagnosis of CDI requires the exclusion of other probable causes of diarrhea in HCT patients and is based on both highly sensitive and highly specific tests to distinguish colonization from infection. The first-line therapy for the first episode is fidaxomicin, while vancomycin is now an alternative agent, and bezlotoxumab, in addition to standard-of-care treatment, has been seen to reduce the risk of recurrent CDI. The treatment of recurrent CDI is based on fidaxomicin or, when not available, on a tapering-pulse regimen with vancomycin; the management of a second or greater rCDI can be challenging and several approaches can be followed.

Several case reports have shown that FMT, which is highly effective in general populations, could be a new valuable strategy for the treatment of rCDI. Further research is required to establish which patients in the allo-HCT population can benefit from FMT to prevent first or recurrent CDI. Currently, a randomized controlled trial is underway (NCT02269150) to assess the efficacy of auto-FMT in this setting. Lastly, new therapeutic approaches for CDI prevention are being explored, including antibiotic neutralization strategies and vaccines.

AUTHOR CONTRIBUTIONS

All authors significantly contributed to this review. Alessandra Mularoni and Elio Castagnola conceived the original idea and edited the manuscript. Davide Lo Porto and Carolina Saffioti contributed equally to the writing of this manuscript. All authors read and approved the final manuscript.

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CONFLICT OF INTEREST STATEMENT

The authors declare no conflict of interest.

DATA AVAILABILITY STATEMENT

Data sharing is not applicable to this article as no new data were created or analyzed in this study.

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

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